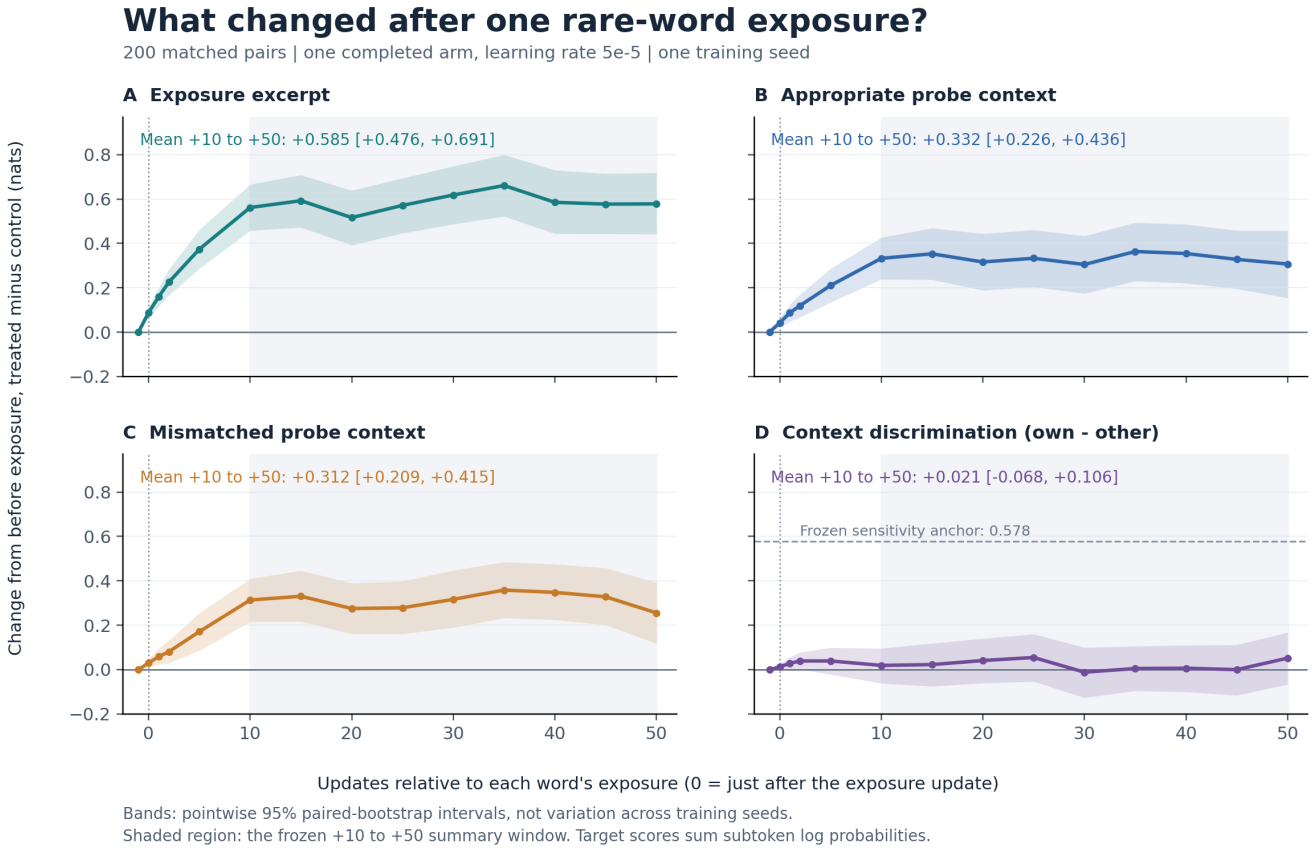
RESEARCH ENGINEERING CASE STUDY | UPDATED 7 NOVEMBER 2026

Rare-word learning after one controlled exposure

Hadi Chamas | Public BabyLM checkpoint

Word probability increased, but persistent context discrimination remained unresolved. The longer forgetting experiment was not launched.

The question was whether one exposure creates a persistent contextual learning signal worth tracking for later forgetting. In one completed continuation run, 200 exposed words were compared with 200 matched controls. The primary appropriate-minus-mismatched contrast was **+0.021 nats**, with a 95% paired-bootstrap interval of **[-0.068, +0.106]**.



Read the contrast, not just the rising curves. The target word became more probable in the exposure excerpt (+0.585), an appropriate probe context (+0.332), and a mismatched context (+0.312). Subtracting the two probe gains leaves an unresolved persistent effect. Scores concern the target word; the exposure result does not establish memorization of the entire sentence.

This is a completed measurement and stopping decision. It is not evidence that contextual learning is impossible, a test of long-term forgetting, or a new training method.

Design, result and stopping rule

Controlled continuation

The public BabyLM-2026 GPT-2 strict-small checkpoint (approximately 98.4M parameters) was pinned to revision `3dc1de10cdad2e54bcc71e9b73d182608d6966f1`. The experiment used a fresh AdamW optimizer. It did not recover the original model's data order or training state.

400 words were matched into 200 pairs by part of speech, exact target subword count and original LT-Swap frequency bin, with approximate context-length matching. Treatment was randomized within pairs. Each treated word appeared once in a cleaned source excerpt; controls were not injected and were measured at matched exposure times. Whole-word absence was checked in the pinned continuation corpus, not certified for all historical pretraining.

After 150 shared warm-up updates, branches ran 50 acclimation updates at constant rates of 5e-5 or 2e-4. The completed 5e-5 branch injected two excerpts per update at updates 201-300 and continued to update 350. Batch size was 16, sequence length 512; short excerpts were masked and the loss retained a fixed denominator of 8,176 nominal target slots.

Measurement fixed before the exposures

Target scores sum subtoken log probabilities. The contextual score subtracts mismatched from appropriate target scores, averaging two probe rows. Each word's change from its own pre-exposure score was calculated, then the matched control change was subtracted. G1 averages these differences over 200 pairs and offsets +10, +15, ..., +50. This is distinct from official LT-Swap full-sentence accuracy.

Intervals use 10,000 paired-bootstrap resamples. Curve bands are pointwise and unadjusted; they do not capture uncertainty across training seeds.

Measure: persistent treated-minus-control change	Nats	95% interval
Target in exposure excerpt	+0.585	[+0.476, +0.691]
Target in appropriate context	+0.332	[+0.226, +0.436]
Target in mismatched context	+0.312	[+0.209, +0.415]
Context discrimination, G1	+0.021	[-0.068, +0.106]

Decision: the lower bound of G1 had to exceed zero to consider a longer chase. It did not. The upper bound also fell below $x^* = 0.578248$, a pre-exposure sensitivity anchor derived from a hypothetical score shift reversing 5% of baseline errors. This anchor is not a validated conversion from mean G1 to repaired decisions; failure of the positive-gain gate is sufficient to stop.

Higher-rate branch: before any rare-word exposure, development NLL rose from 4.66354 to 4.78614 (+2.63%), exceeding the 2% guard. It is not evaluable for acquisition. A development-loss guard is not proof of numerical divergence.

Excluding flagged-source pairs gives G1 = +0.054 [-0.040, +0.144]; an exposure-event bootstrap gives [-0.075, +0.110]. The secondary +2.5-point directional accuracy change has no reported interval. A nominally positive pointwise contrast at +2 does not replace the frozen persistent endpoint.

Engineering judgment and limits

Stop this research line here. The result supports a reproducible case study; it does not justify the planned long chase or a replay intervention.

Caught before further spending

Formatting: 329/400 candidate excerpts had markup and/or spaced punctuation (160 treated, 169 controls). Cleaning removed detected patterns without changing words, assignments or times. Residual source-domain differences remain.
Threshold: a frequency-gap comparison mixed frequent single-token with rare multi-token words. It was replaced with the pre-exposure anchor, whose interpretation is explicitly limited.
Staging: a pre-exposure stability check and a short acquisition gate prevented the two branches from automatically entering the expensive long chase.

Costs and recovery

The successful A100 Stage A invocation took **425.50 seconds** and an estimated **0.795 Colab units** at the supplied 6.73 units/hour. The earlier profile separately used about 0.204 estimated units. Actual billing, idle time, failed launches and earlier research are excluded. A previous oversized notebook stalled at launch and was repackaged. Numerical preflight tests had not covered browser startup.

The earlier short A100 profile reported exact save/reload replay; synthetic CPU tests covered interrupted driver execution. The package now rebuilds all plotted statistics on CPU from saved numerical observations, with no model loading or new training. These checks do not guarantee every browser or hardware failure mode.

Limits that change the interpretation

One seed and one completed low-rate arm; one exposure per treated word; 397/400 words have only one distinct positive context; fresh-optimizer continuation; no certified historical absence; no interval across training seeds. Shared-model dependence and source-domain differences remain. The study cannot rule out a smaller effect, other stable rates, or learning after repeated exposure, and it never measured long-term forgetting.

Project and sources

Hadi Chamas: project direction, experiment execution and stopping decisions. The project used AI assistance for code, analysis and design review. The BabyLM community supplied the checkpoint; Algayres and colleagues supplied LT-Swap.

To rebuild: install **requirements.txt**, then run **python reproduce.py**. The public repository includes numerical observations, the frozen protocol, analysis code, provenance and figures: github.com/Hc1012/rare-word-pulse. Updated 7 November 2026; scientific results unchanged.

Sources

[1] BabyLM public GPT-2 strict-small checkpoint; immutable revision above.
[2] Algayres et al. (2025). LongTail-Swap. DOI: 10.18653/v1/2025.findings-emnlp.601.
[3] Chang & Bergen (2022). Word Acquisition in Neural Language Models. DOI: 10.1162/tacl_a_00444.
[4] Chang, Tu & Bergen (2024). Characterizing Learning Curves During Language Model Pre-Training. DOI: 10.1162/tacl_a_00708.
References [3-4] provide related acquisition/learning-curve work; this case study does not reproduce their experiments. Raw result provenance and hashes are in the accompanying package.